

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

WIRELESS COMMUNICATION

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions carry marks as indicated.
 - (2) All questions are compulsory.
 - (3) Assume suitable data wherever necessary.
 - (4) Illustrate your answers with help of neat sketches wherever necessary.
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1.
 - (a) Demonstrate the concept of Cell splitting in detail for the improvement of Wireless Communication System. 5(CO1)
 - (b) Consider a single high-Power transmitter that can support 40 voice channels over an area of 140 Km^2 , With the available spectrum. If this area is equally divided into seven smaller areas (cells), each supported by lower power transmitter so that each cell supports 30% of the channels, then determine :
 - (i) Coverage area of each cell.
 - (ii) Total no. of voice channels available in cellular system.
 - (iii) Comment on the result obtained. 5(CO1,4)
 2.
 - (a) Assume if a transmitter produces 50W of power, express the transmit power in units of dBm and dBW. If 50 W is applied to a unity gain antenna with a 900 MHz carrier frequency, find the received power in dBm at a free space distance of 100m from the antenna, also justify the analytical expression by computing the received power at 10 Km. 7(CO2)
 - (b) List and explain the reasons for fading in wireless communication. 3(CO2)

3. Write note on any two algorithms for adaptive equalization and then compare them.
10(CO3)
4. (a) Identify the multiple access technique which were used in 1G and 2G wireless communication and compare them. 6(CO3)
- (b) If GSM uses a frame structure where each frame consists of eight time slots and each time slot contains 156.25 bits and data is transmitted at 270.833 kbps in the channel, find :
- (a) The time duration of a bit,
- (b) The time duration of a slot,
- (c) The time duration of a frame, and
- (d) How long must a user occupying a single time slot wait between two successive transmissions ? 4(CO4)
5. Demonstrate architecture of GSM in detail with the help of block diagram. 10(CO5)
6. Illustrate the concept of MIMO communication with its advantages and disadvantages. 10(CO5)

